PARTICIPANT BLUEPRINT & CATALOGUE

IPL HACKATHON: PROBLEM STATEMENTS

Algorithmic decision-making systems and scenario-based technical challenges

Time Available: 5–6 Hours
Format: Working Prototype Only
Decks / PPTs: Strictly None

> **Core Design Expectation:** Build systems that make defensible decisions, explain the evidence behind recommendations, and adapt when constraints shift—rather than static display dashboards.

5-HOUR PROTOTYPE BLUEPRINT: HOW TO SCOPE & DELIVER

WHAT NOT TO BUILD. DO NOT SPEND TIME HERE

- **No User Auth or Signups:** Hardcode user sessions or bypass login screens completely.
- **No Production Database Setups:** Avoid provisioning cloud DBs. Keep state in-memory or in local mock JSON/CSV files.
- **No Slide Decks (No PPT):** Presentation decks are prohibited; judging is 100% demo-driven.
- **No Complex Multi-Page UI:** Avoid nested routing or mobile-responsive CSS perfection.

WHAT TO BUILD HIGH VALUE DELIVERABLES

- **Single-View Reactive Dashboard:** Build with Streamlit, Gradio, Next.js, or simple web interfaces.
- **Scenario-Driven Mock Data:** Seed inputs using the provided *Target Demonstration Scenario*.
- **Traceable Decision Engine:** Showcase clear deterministic or AI reasoning behind flags and actions.
- **Full AI Acceleration:** You are actively encouraged to leverage LLMs and AI code generation.

RECOMMENDED 3-STEP DEMONSTRATION PATTERN

STEP 1: INGEST & TRIGGER

Load sample orders, inject an unplanned machine outage, or feed contradictory claim documents live on screen.

STEP 2: ENGINE PROCESSING

Show automated logic detecting duplicate billings, constraint rebalancing, or conflicting field logs in real time.

STEP 3: ACTION & EVIDENCE

Display the recommended outcome (approve, reschedule, escalate) backed by transparent confidence and citation logs.

JUDGING EVALUATION CRITERIA

1. **Correctness & Constraints:** Handles edge cases, missing data, and conflicts.
2. **Reasoning & Explainability:** Clear justification and traceable evidence behind decisions.
3. **Adaptability:** Dynamic response when inputs or operating conditions shift.
4. **Usability:** Clear, actionable outputs demonstrated live during evaluation.

1. Multi-Source Business Discrepancy Investigator

Business Operations • Data Engineering • Decision Support

Background & Objective: Organizations receive purchase orders, invoices, shipment receipts, and payment confirmations across fragmented systems. Discrepancies emerge from partial deliveries, duplicate billings, differing item naming conventions, and timing delays. Build an explainable system that ingests multi-source transaction records, reconstructs the true operational state of each transaction, distinguishes harmless variances from costly errors, and explains the supporting evidence.

KEY DELIVERABLES

- Match interdependent records even when keys, descriptions, quantities, or dates differ slightly.
- Detect duplicate charges, partial deliveries, omitted records, and currency or amount mismatches.
- Present traceable confidence levels and explicit evidence trails behind every flagged discrepancy.
- Prescribe next operational steps (Approve, Request Clarification, Escalate, or Place on Hold).
- Appropriately categorize ambiguous entries rather than generating fabricated conclusions.

Target Demonstration Scenario: A purchase order is partially fulfilled, the supplier generates duplicate invoices, and payments split across transfers. Reconstruct total transactions, compute true balances, and flag ambiguities.

2. Production Disruption Recovery Planner

Manufacturing • Dynamic Scheduling • Operations Research

Background & Objective: Modern manufacturing links job orders, machine capacity, labor shifts, and component availability. Unexpected machine downtime or shipment bottlenecks make baseline plans infeasible, causing cascading delays and heavy penalties. Build a production recovery engine that ingests dynamic operational disruption events and autonomously recalculates feasible schedules balancing order priority and machine compatibility.

KEY DELIVERABLES

- Model order work orders as sequence-dependent operational stages with shift/material bounds.
- Map machine compatibility, throughput rates, maintenance downtime, and delivery deadlines.
- Pinpoint immediately affected downstream work orders whenever unplanned interruptions occur.
- Formulate and evaluate alternative production schedules against clear objective criteria.
- Quantify and explain operational trade-offs (tardiness, idle machine hours, changeovers).

Target Demonstration Scenario: Primary machinery suffers a 4-hour breakdown while an input shipment delays, with critical orders due by end-of-day. Produce a feasible schedule update illustrating affected commitments.

3. Evidence-Based Claim Verification & Decision Support

Document Intelligence • Rule Engines • AI-Assisted Review

Background & Objective: Warranty, transit-damage, and insurance claims involve semi-structured forms, photographic proofs, invoices, timestamps, and legal policies. Reviewing packets manually is time-intensive, while blind automation risks severe oversight. Develop an evidence-evaluation platform that cross-references unstructured/structured proofs against regulatory policies to determine verified facts, overt contradictions, and missing items.

KEY DELIVERABLES

- Extract and normalize entities across incident filings, item receipts, and documentation.
- Cross-validate dates, reported damages, item models, policy clauses, and declared values.
- Implement structured policy verification rules to uncover discrepancies and false statements.
- Highlight contradictory data points with explicit references to the primary input source.
- Generate transparent recommendations (Approve, Request Supplemental Proof, Human Audit).

Target Demonstration Scenario: Damaged equipment claims are filed post-delivery where repair receipts, purchase timestamps, and photographic condition contradict declared values. Identify discrepancies and prompt for missing proofs.

4. Adaptive Energy Demand & Load Scheduler

Energy Optimization • Demand Forecasting • Smart Scheduling

Background & Objective: Commercial buildings and industrial facilities navigate fluctuating grid tariffs, limited peak-load thresholds, and volatile onsite renewable generation. Simultaneously activating loads triggers severe tariff penalties or infrastructure trips. Develop an adaptive scheduling engine allocating power across critical and deferrable machinery based on tariff cycles and generation forecasts.

KEY DELIVERABLES

- Forecast short-term demand trends using historical consumption profiles or simulated datasets.
- Distinguish mandatory non-interruptible loads from flexible, time-shiftable operational devices.
- Generate optimized device dispatch schedules adhering strictly to maximum peak thresholds.
- Dynamically adapt schedule allocations upon real-time solar drops or spot price spikes.
- Quantify cost efficiencies, peak-shaving benefits, and trade-offs made across operational plans.

Target Demonstration Scenario: Onsite solar drops unexpectedly during peak tariff hours. Re-allocate power immediately to protect mission-critical equipment while shifting deferrable cycles and logging cost impact.

5. Field Operations Conflict Resolver

Offline-First Systems • Distributed Sync • Data Integrity

Background & Objective: Technical field personnel frequently operate in environments with intermittent connectivity, updating inventory, work tickets, diagnostic notes, and sign-offs locally. When multiple technicians modify shared tickets concurrently while disconnected, default timestamp overwrites corrupt records or destroy history. Build an offline-capable field management architecture with deterministic conflict-resolution rules.

KEY DELIVERABLES

- Persist and queue operational updates safely within local client storage while disconnected.
- Intelligently reconcile queued changes back to central systems once network access resumes.
- Detect stale edits, overlapping field alterations, and conflicting parts consumption logs.
- Execute deterministic automated resolution protocols according to explicit business logic.
- Preserve complete audit trails for unresolved conflicts requiring manual supervisor review.

Target Demonstration Scenario: Two technicians update maintenance progress on the same unit offline, logging conflicting replacement components and status updates. Synchronize without data loss or overwriting.

6. Intelligent Product Returns & Recovery Engine

Reverse Logistics • Operational Optimization • Decision Systems

Background & Objective: Handling retail merchandise returns requires evaluating return fraud, product conditions, refurbishment expenses, warranty liabilities, and secondary market recovery margins. Without algorithmic decision-support, items are routed sub-optimally—yielding inflated shipping costs or avoidable disposal write-downs. Build an automated reverse-logistics decision engine that maximizes economic recovery.

KEY DELIVERABLES

- Ingest return assessments (condition reports, defects, categories, warranty entitlements).
- Dynamically model economic outcomes: restock as new, repair/refurbish, harvest, recycle, or scrap.
- Factor facility throughput bottlenecks (technician hours, refurbishment bench capacity).
- Formulate clear explanations justifying the monetary and operational merit of each routing.
- Isolate ambiguous damage reports or inconclusive cases for secondary physical assessment.

Target Demonstration Scenario: Ten returned devices arrive with varying faults, warranty statuses, and repair costs, but repair bench capacity is limited to four. Prioritize cases and assign optimal disposition.

7. Public Grievance Root-Cause & Action Prioritizer

Civic Tech • Natural Language Processing • Action Queues

Background & Objective: Civic administration helplines receive high-volume incident reports with fragmented descriptions, varied terminology, and multiple languages. Simple complaint tallies fail to distinguish concentrated single-point emergencies from broad systemic defects. Build an intelligent triage platform that aggregates semantically similar complaints, clusters incident hotspots, and outputs prioritized remediation queues.

KEY DELIVERABLES

- Group semantically equivalent reports despite varied phrasing, slang, or language choice.
- Distinguish single-incident complaint storms from recurring community infrastructure issues.
- Compute dynamic priority weights using severity, affected citizen density, duration, and safety.
- Provide transparent justification and primary grievance excerpts for prioritized work queues.
- Track whether municipal team interventions result in measurable drops in repeat reports.

Target Demonstration Scenario: Ingest an unorganized stream of municipal complaints containing duplicates, varied descriptions, and acute hazards. Cluster reports and output a transparently prioritized action queue.

8. Inventory Shortage Early-Warning & Substitution Planner

Supply Chain • Predictive Analytics • Constraint Optimization

Background & Objective: Retail networks and light manufacturers frequently face stockouts despite having adequate aggregate inventory, due to geographic imbalances, supplier delays, and rigid component requirements. Build a predictive stockout prevention engine that forecasts impending bottlenecks, assesses compatibility rules, and recommends automated warehouse transfers, safety stock adjustments, or component substitutions.

KEY DELIVERABLES

- Forecast potential stockouts across distributed warehouse nodes using replenishment lead times.
- Identify functional component or SKU substitutes based on strict engineering compatibility.
- Account for pre-committed order stock to avoid false-availability recommendations.
- Model and compare financial impacts between express reorders, warehouse transfers, or delays.
- Output actionable dispatch and substitution orders with full auditability and cost reasoning.

Target Demonstration Scenario: A core component shipment is delayed while multiple orders compete for dwindling stock across regional hubs. Recommend optimized stock transfers, reorders, or approved part substitutions.

Final Demonstration Reminder: Judges will evaluate your live working prototype based on your engine's logic, edge-case handling, and explainability using your scenario dataset. Avoid spending time on cosmetic extras—build core decision power.